

REPRODUCIBILITY

How to Run the Code

1. Install required tools

`sudo apt update`

`sudo apt install -y audacity ffmpeg python3 python3-pip`

2. Install Python libraries

`pip3 install numpy librosa matplotlib soundfile pydub mutagen demucs`

3. Prepare audio files

- Put your input .wav file in the project folder
- Use Audacity to:
 - Load the audio
 - Separate into: vocals, drums, bass, others
 - Export to:

`output/stems/`

4. Compress stems

- Use FFmpeg or Audacity to compress each stem
- Save as:

`vocalscompress.wav`

`drumscompress.wav`

`basscompress.wav`

`othercompress.wav`

5. Run reconstruction

`python3 reconstruct.py`

Output:

`reconstructed_mix.wav`

6. Show bitrate

`python3 show_bitrate.py`

This will display bitrate and sample rate for each stem.

7. Analyze audio

```
python3 analyze_single.py
```

This shows:

- Waveform
- RMS energy
- Spectrogram

How to Reproduce Results

To reproduce the results in the report, follow the same pipeline:

1. Use the same input audio (music or speech)
2. Perform separation into 4 stems
3. Apply the same compression method for all stems
4. Run reconstruction
5. Compare:
 - Original vs reconstructed audio (by listening)
 - Bitrate per stem
 - Waveform and spectrogram

Commands Summary

```
# Run reconstruction
```

```
python3 reconstruct.py
```

```
# Check bitrate
```

```
python3 show_bitrate.py
```

```
# Analyze audio
```

```
python3 analyze_single.py
```

Configurations

- Audio format: .wav
- Sample rate: keep original (no resampling)
- Channels: stereo (2 channels)
- Compression: same method applied to all stems for fair comparison

